

MODEL PERFORMANCE COMPARISON

1.SVM/SVR- Kernel Experiment

Kernel type	Parameter	R2 score	Status
Linear	Kernel=linear	0.895077	Best
RBF- non linear	Kernel=rbf	-0.05731	Need scaling
Standardisation		-0.05731	
Poly	Kernel=poly	-7.65328	Flop
Sigmoid	Kernel=sigmoid	-0.05700	
C value	C=1000	-0.05700	No improvement

Observation:

In SVR, scaling is not required for the Linear kernel. Since RBF, Poly, and Sigmoid are distance-based kernels, the R2 score becomes negative without Standardisation. The Linear kernel gave the best score of 0.895.

2.Decision Tree Regression Experiment

Criterion	Splitter	R2score	Status
Default		0.92405	Good
friedman_mse	best	0.931135	Highest-Best model
friedman_mse	random	0.792325	Low
Squared_error	best	0.894838	Good

Conclusion: In Decision Tree, criterion=friedman_mse and splitter=best is the overall best model with a score of 0.931135. This can be taken as the final model as it outperformed all SVR kernels.

Final Model: Decision Tree (friedman_mse + best) = 0.931135